PR. 5 — SMART OUTCOME PREDICTOR

Part A: Conceptual Foundation (Theory) · Only the 5 questions from your project brief

1. What are Ensemble Learning Methods and why are they effective?

Ensemble learning combines predictions from multiple machine-learning models to produce a stronger result. It is effective because different models can make different errors, so combining them can improve accuracy, stability, and generalization.

2. Explain the difference between Bagging, Boosting, and Voting.

Bagging trains models independently on different bootstrap samples and combines their predictions. Boosting trains models sequentially, with later models focusing on earlier errors. Voting combines predictions from multiple models using majority voting or averaged probabilities.

3. What is Bias–Variance Trade-off, and how do ensemble methods address it?

Bias is error from a model being too simple, while variance is sensitivity to changes in training data. Bagging often reduces variance by averaging models, while boosting can reduce bias by learning patterns missed by earlier models. The best balance depends on the dataset.

4. Explain Voting Classifier and Stacking Ensemble.

A Voting Classifier combines predictions from multiple classifiers: hard voting selects the majority class, while soft voting averages predicted probabilities. Stacking uses predictions from base models as inputs to a second model called a meta-learner, which learns how to combine them.

5. Compare AdaBoost and XGBoost conceptually.

AdaBoost gives more attention to training examples that previous weak learners misclassified. XGBoost is an optimized gradient-boosting method that builds trees sequentially and uses regularization to help control overfitting. Both combine learners to improve predictive performance.

Note: This PDF intentionally contains only Part A (Conceptual Foundation) from your PR. 5 document. Dataset tasks, algorithms implementation, evaluation metrics, and extra viva questions are not included.